

# Experimental Materials Characterization

Synthesis of Carbon Fibers with Carbon Nanotubes  
via Electrospinning and Chemical Vapor Deposition

Sofía Carolina Quintanilla Galindo    Santiago Guillermo Mejía Torres  
Gaby de la Garza González    Gabriel Ortiz Valdivieso    Bernardo Lugo Albor

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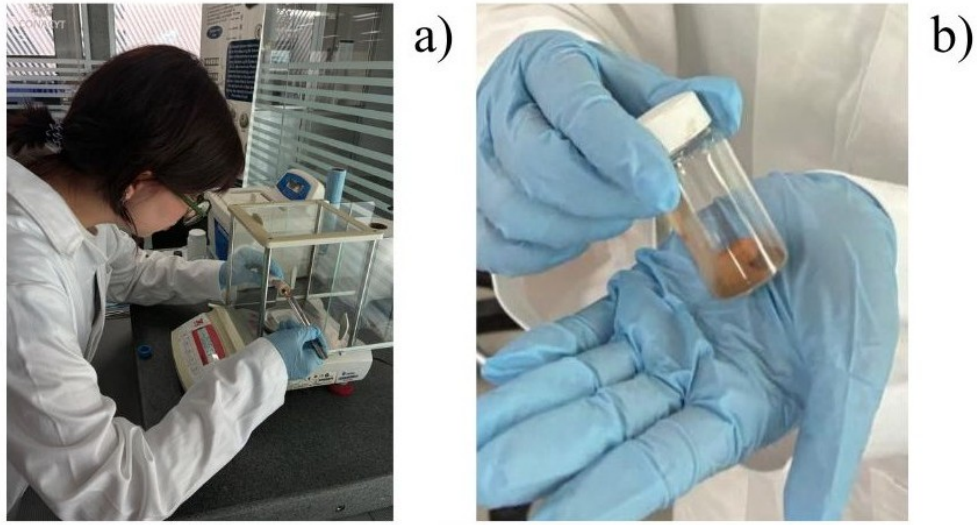
## Abstract

This report details the synthesis of carbon fibers embedded with carbon nanotubes (CNTs) using electrospinning and chemical vapor deposition (CVD). A solution of polyacrylonitrile (PAN) dissolved in dimethylformamide (DMF) was prepared with iron nanoparticles (Fe NPs) as catalysts. The solution was processed by electrospinning under controlled parameters — voltage, flow rate, needle-to-collector distance, and environmental conditions — producing a stable Taylor cone and a continuous nanofibrous mat. The mat was subsequently subjected to pyrolysis and CVD at 900 °C in an inert nitrogen atmosphere, during which the iron nanoparticles promoted CNT growth on the fiber surface. The results demonstrate the feasibility of integrating electrospinning and CVD for the fabrication of nanostructured carbon materials with potential applications as electrodes and sensors.

## Solution Preparation

The objective of this stage was to prepare a homogeneous solution of DMF (solvent) with PAN and iron nanoparticles (Fe NPs) as solutes, suitable for electrospinning and subsequent CVD/pyrolysis.

First, 0.6053 g of Fe NPs (80% by mass relative to PAN) were weighed on a Pioneer balance (sensitivity 0.0001 g) and placed in a glass vial with a magnetic stir bar.

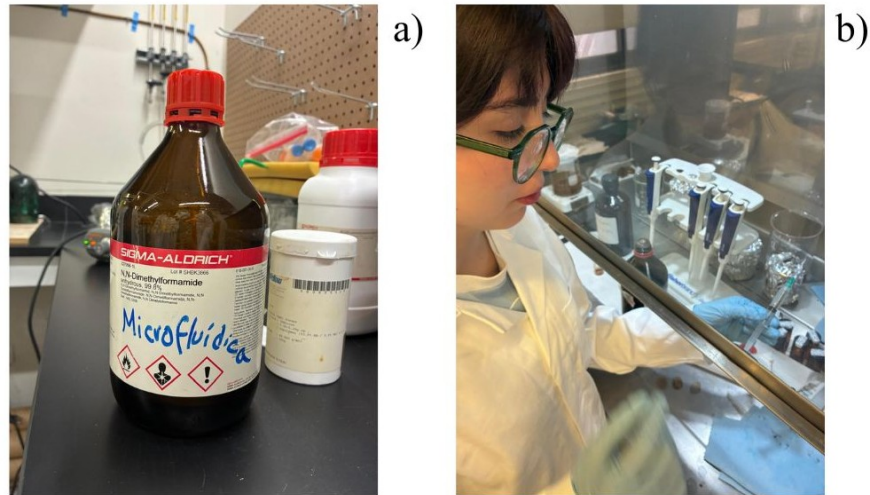


**Figure 1:** (a) Weighing of iron nanoparticles on a Pioneer balance. (b) Iron nanoparticles with magnetic stir bar.

DMF was then added as the solvent. It is a polar organic solvent widely used in electrospinning due to its high boiling point and ability to dissolve PAN. A volume of 4 mL was used, converted to grams as follows:

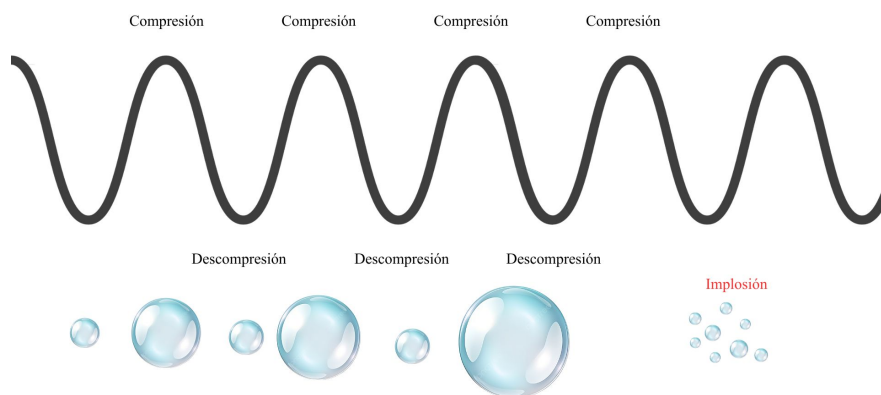
$$V \cdot \rho = 4 \text{ mL} \times 0.944 \frac{\text{g}}{\text{mL}} = 3.776 \text{ g.}$$

This step was performed inside a fume hood, as DMF is toxic by inhalation.



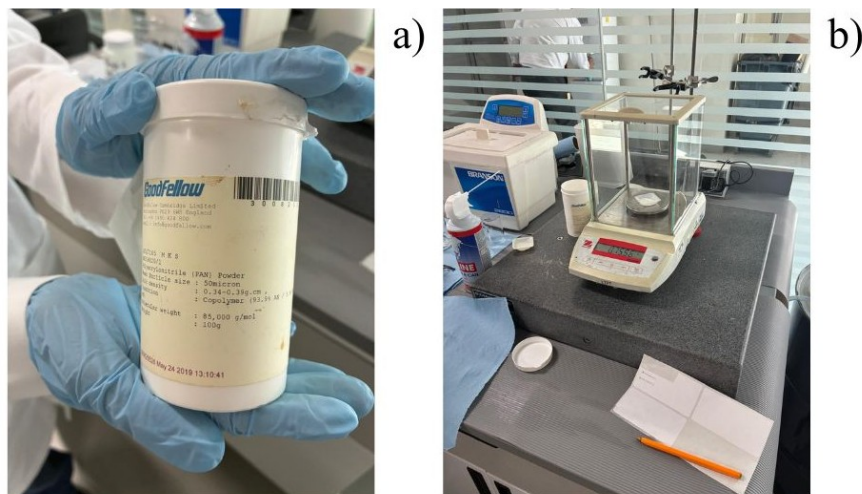
**Figure 2:** (a) *N,N*-Dimethylformamide. (b) DMF extraction inside the fume hood.

The suspension was then placed in an ultrasonic bath for 20 minutes to break up agglomerates and improve particle dispersion. The bath generates high-frequency waves that produce cavitation: microscopic bubbles collapse and release localized high-energy pulses, homogenizing the mixture.



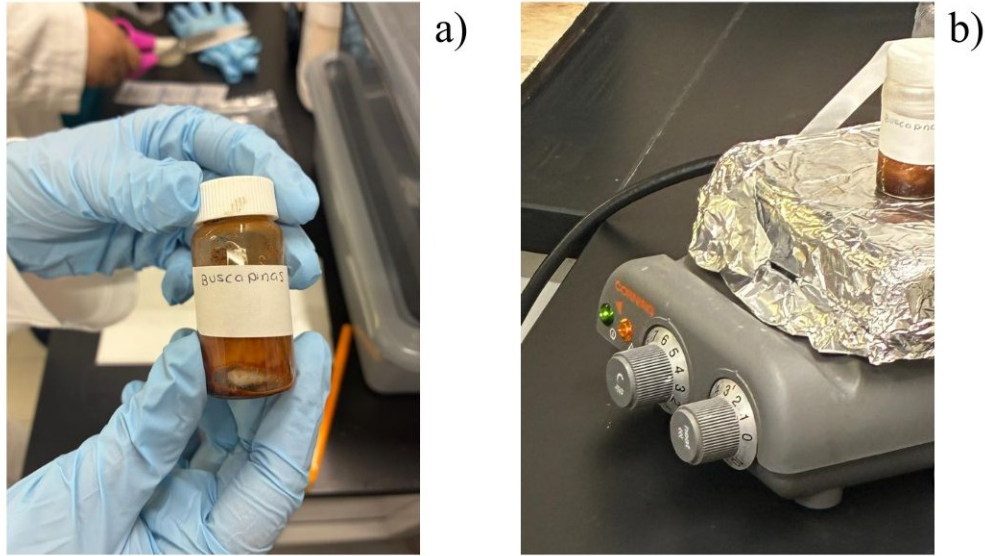
**Figure 3:** Schematic of the cavitation process in the ultrasonic bath.

Subsequently, 0.7555 g of PAN were weighed and added to the DMF suspension. PAN provides the viscosity necessary for fiber formation during electrospinning.



**Figure 4:** (a) Polyacrylonitrile (PAN). (b) Weighing of PAN on the Pioneer balance.

The solution was sealed with stretch film to prevent solvent evaporation and placed on a Corning magnetic hot plate for 12 hours to ensure complete dissolution of PAN.

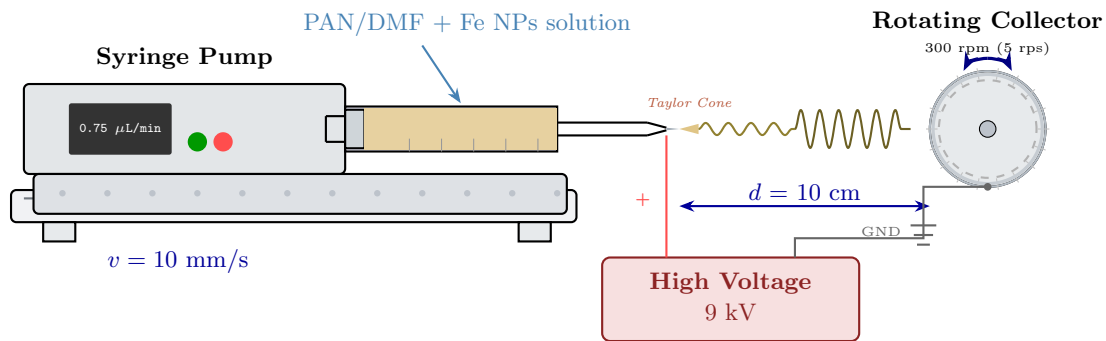


**Figure 5:** (a) PAN/DMF/Fe NP solution. (b) Solution on the magnetic hot plate stirrer.

## Electrospinning

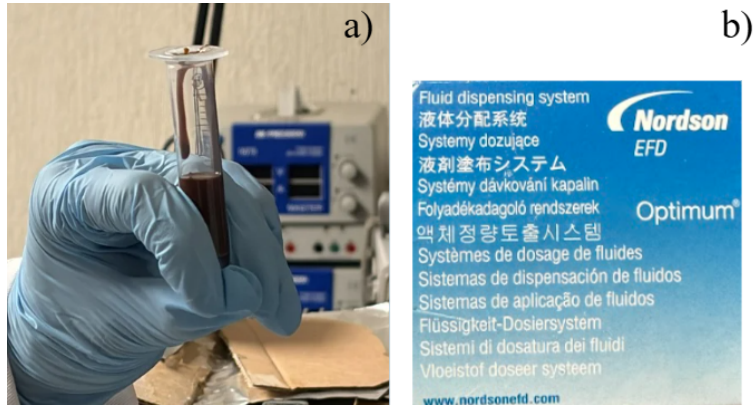
Electrospinning converted the PAN/DMF/Fe NP solution into a nanofibrous mat, which served as the structural support for CNT growth in the subsequent CVD stage.

Figure 6 shows a schematic of the electrospinning system. The solution is fed through a syringe pump, a high voltage is applied between the metallic needle and a rotating cylindrical collector, and the resulting electric field deforms the droplet at the needle tip into a Taylor cone, from which a polymer jet is ejected and deposited as a fiber mat.

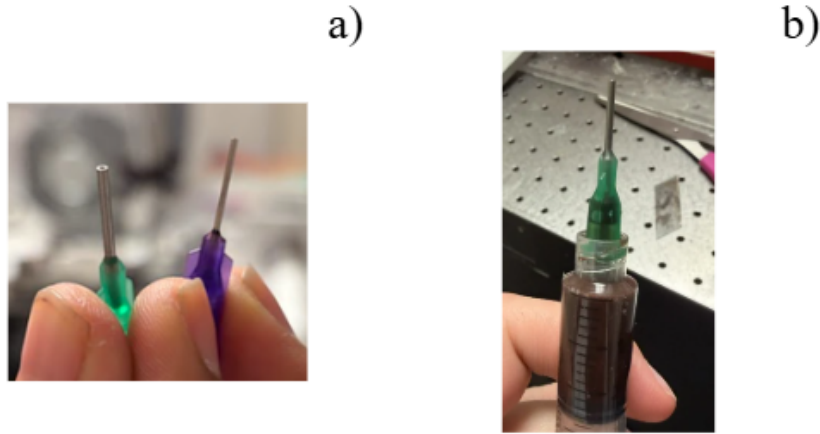


**Figure 6:** Schematic of the electrospinning system. Main components: syringe pump, PAN/DMF + Fe NP solution (yellow), metallic needle connected to the high-voltage positive terminal, and grounded rotating cylindrical collector.

The syringe was loaded without air by opening the plunger, filling with solution while blocking the outlet, and gently closing it. The metallic needle was connected to the syringe and the collector was covered with parchment paper. The pump was set to  $0.75 \mu\text{L}/\text{min}$  and primed at  $0.15 \mu\text{L}/\text{min}$  until a droplet formed at the needle tip, purging any residual air.

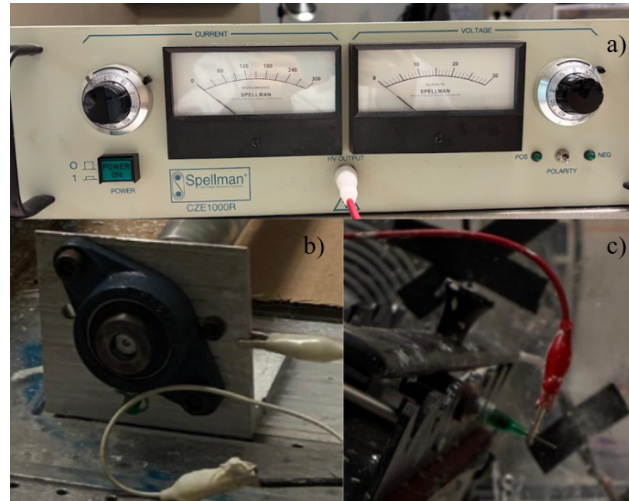


**Figure 7:** Syringe loading process. (a) Filled syringe. (b) Syringe model used.



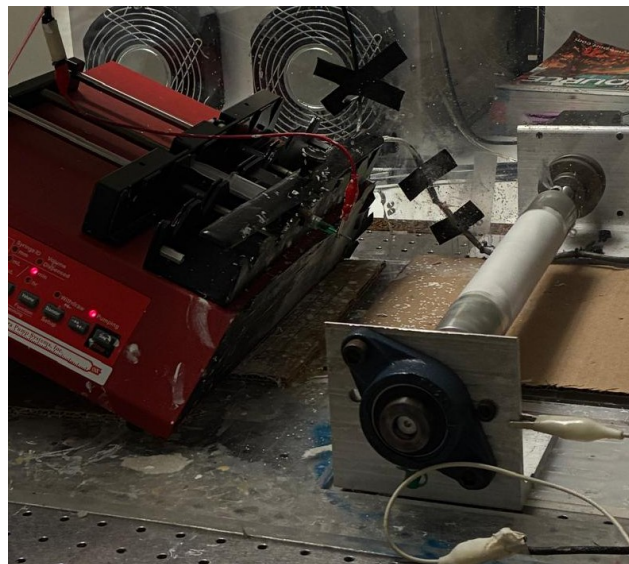
**Figure 8:** Needles available in the laboratory. (b) Syringe with the needle selected based on solution viscosity.

The rotating collector was set to 5 rps and the voltage was ramped from 0 to 9 kV using a Spellman CZE1000R source. Due to the high viscosity of the solution, the lateral translation stage had to be bypassed, as sedimentation at the connections prevented proper flow. This reduced spatial control over fiber thickness and mat uniformity.



**Figure 9:** *Electrospinning voltage source. (a) Spellman CZE1000R model. (b) Positive connection at needle. (c) Ground connection at collector.*

Once stable operating conditions were established, the Taylor cone formed at the needle tip and a continuous polymer jet was directed toward the collector center. The process ran for 2 hours with continuous monitoring of the Taylor cone stability; minor voltage adjustments ( $\pm 0.5$  kV) and needle cleaning were performed whenever dripping was observed.



**Figure 10:** *Final electrospinning setup with operational modifications.*



**Figure 11:** *Nanofibrous mat deposited by the electrospinning process.*

Relative humidity was maintained below 60% throughout the process using a Friedrich dehumidifier, as excess humidity causes surface pores in the fibers due to solvent condensation during jet travel.



**Figure 12:** *Friedrich dehumidifier used for environmental control.*

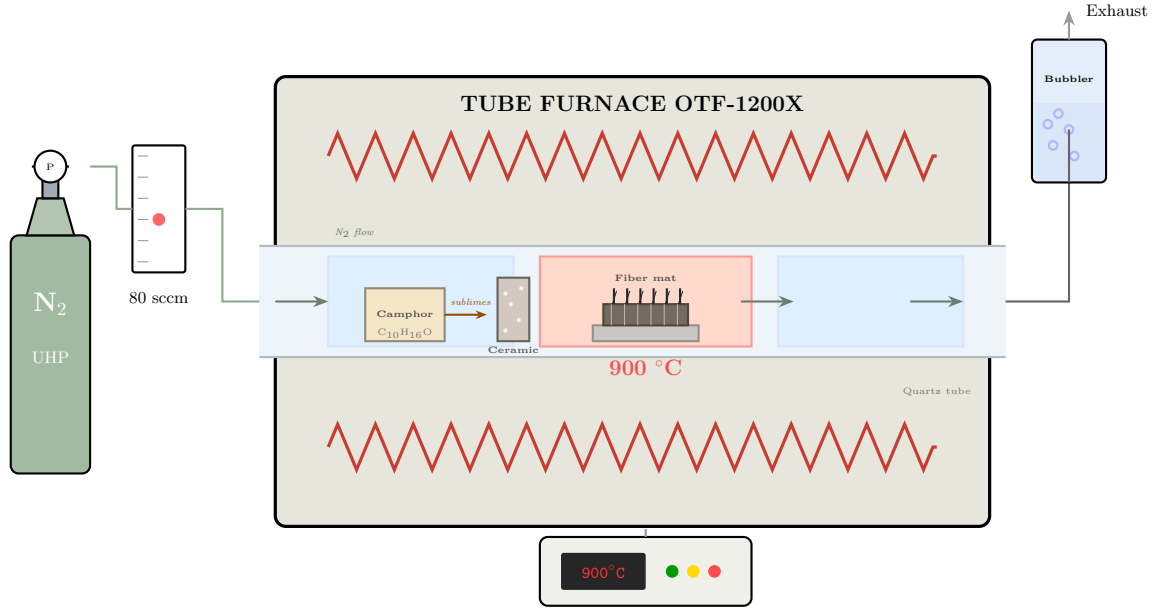
Table 1 summarizes the most frequent issues encountered during the process and the corrective actions applied.

**Table 1:** *Common issues during electrospinning and corrective actions taken.*

| Observed problem                                     | Probable cause                              | Corrective action                            |
|------------------------------------------------------|---------------------------------------------|----------------------------------------------|
| Dripping at needle tip without Taylor cone formation | Insufficient voltage or excessive flow rate | Increase voltage by 0.5 kV; reduce flow rate |
| Multiple jets                                        | Excessive voltage                           | Reduce voltage by 0.5 kV                     |

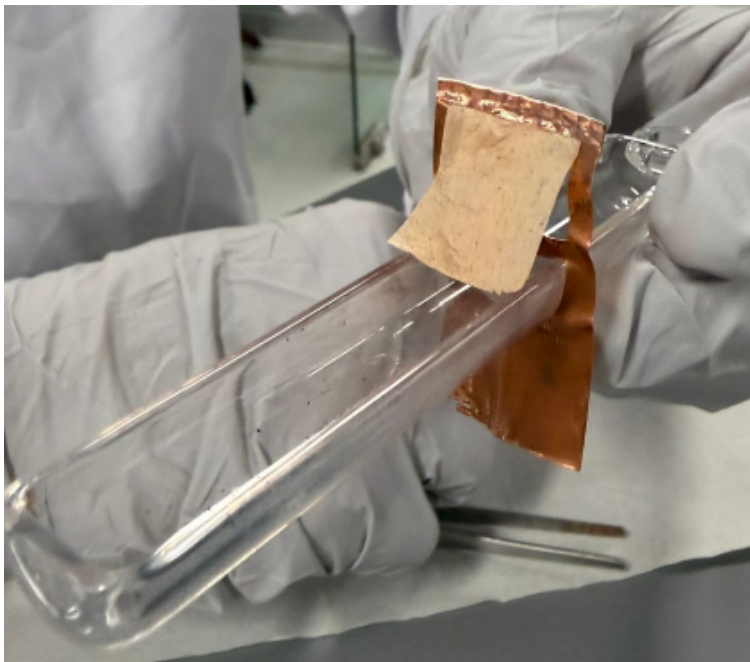
## Pyrolysis and Chemical Vapor Deposition

After electrospinning, the nanofibrous mat was processed by pyrolysis and CVD to carbonize the PAN matrix and grow CNTs on the fiber surface using camphor as a carbon precursor and the Fe NPs as catalysts.



**Figure 13:** Schematic of the pyrolysis and CVD system. The OTF-1200X tube furnace with a quartz tube is fed with nitrogen; camphor sublimates and is carried by the gas flow over the fiber mat, where Fe NPs catalyze CNT growth.

A rectangular section was cut from the electrospun mat using its protective parchment paper backing to avoid damage. A larger copper foil was then cut with a rectangular slot to anchor the mat and a semicircular recess to hold the quartz crucible.



**Figure 14:** Quartz crucible arrangement with the fiber mat mounted on the copper foil.

The CVD process was carried out in a horizontal tube furnace (MTI Corporation, model OTF-1200X-S-II) with a 2-inch-diameter quartz tube. The assembly (Figure 14) was placed at the center of the tube. Camphor (Sigma-Aldrich, 96°C,  $13.0456 \pm 0.0001$  g) was placed in a quartz crucible separated from the fiber mat by a ceramic cylinder to distribute the vapor uniformly. Camphor sublimation was driven by a thermal tape regulated with a variable autotransformer (Staco Energy Products, model 3PN1010B) placed outside the quartz tube.



**Figure 15:** Component arrangement inside the tube furnace: (a) camphor crucible, (b) ceramic separator, (c) electrospun fiber mat sample.

The system was sealed with O-rings and steel clamps. Ultra-high-purity nitrogen ( $N_2$ ) was introduced at 80 sccm to purge atmospheric oxygen; correct sealing was verified by continuous bubbling in the water-trap outlet.



**Figure 16:** *Nitrogen tank used to maintain an inert atmosphere throughout the CVD process.*

The furnace temperature ramp was set to reach 900 °C in 90 minutes. The thermal tape was activated 60 minutes into the ramp so that effective camphor evaporation coincided with the furnace reaching its target temperature. The system was held at 900 °C for 90 minutes under continuous N<sub>2</sub> flow, during which PAN pyrolysis and CNT growth occurred simultaneously. The furnace was then allowed to cool naturally under N<sub>2</sub> flow to prevent sample oxidation. The total process time, including cooling and sample retrieval, was approximately 8.5 hours.



**Figure 17:** *Full experimental setup for the CVD process, operated inside a fume hood.*

All high-temperature and gas-handling steps were performed inside a fume hood, with the outlet bubbler

providing continuous flow verification.

## Results

Carbon fibers with surface-grown CNTs were successfully produced. Starting from the PAN/DMF/Fe NP solution, a fiber mat was deposited at 9 kV, 0.75  $\mu\text{L}/\text{min}$ , and a 10 cm needle-to-collector distance. Pyrolysis and CNT growth were performed at 900  $^{\circ}\text{C}$  for 90 minutes in  $\text{N}_2$  atmosphere with camphor as the carbon precursor.



**Figure 18:** *Final synthesis product: carbon fiber mat with carbon nanotubes.*

The product (Figure 18) exhibits macroscopic indicators consistent with CNT growth: black coloration and characteristic surface texture. However, compositional non-uniformity was observed and attributed primarily to the high viscosity of the PAN solution, which prevented stable operation of the lateral translation stage and led to non-uniform fiber deposition during electrospinning. This likely affected the homogeneity of CNT growth during CVD.

The results are consistent with the methodologies reported by Zhang et al. (2011) and De Volder et al. (2010), confirming the viability of integrating electrospinning and CVD for carbon electrode fabrication. Future work should address the nanoparticle concentration in solution, camphor sublimation kinetics, and the furnace temperature ramp profile. Structural characterization of the product — including SEM, Raman spectroscopy, and TEM — remains necessary to confirm CNT morphology and assess composition.

## References

1. M. De Volder, S. H. Tawfick, S. J. Park, D. Copic, Z. Zhao, W. Lu, and A. J. Hart, “Diverse 3D microarchitectures made by capillary forming of carbon nanotubes,” *Advanced Materials*, vol. 22, pp. 4384–4389, 2010.
2. L. Zhang, T. Shi, S. Xi, D. Liu, Z. Tang, X. Li, and W. Lai, “Carbon nanotube integrated 3-dimensional carbon microelectrode array by modified SU-8 photoresist photolithography and pyrolysis,” *Thin Solid Films*, vol. 520, pp. 1041–1047, 2011.